SAINT MICHAEL COLLEGE OF CARAGA

Michaelinian Quest: A Story-driven Role-playing Game Exploring the Core Values of Saint Michael College of Caraga

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CHAPTER 1

INTRODUCTION

1.2 Project Context

In today's digital learning environment, educational institutions face increasing challenges in engaging students in meaningful, value-based education. Saint Michael College of Caraga (SMCC) strives to uphold its core values. Even though they are socially responsible, missionary, committed, and competent, traditional methods like literature, posters, and classroom orientations often do not engage students or encourage deep moral questioning. As digital natives, students connect better with interactive and tech-based learning experiences that mix engagement with moral growth. In online learning, students give feedback more effectively in these interactive experiences. The rising need for new teaching methods supports the development of an interactive digital platform that reflects SMCC's values. In digital learning, students provide feedback more effectively in interactive, technology-driven learning experiences that integrate engagement with moral formation. The growing demand for innovative educational approaches justifies the creation of an interactive digital platform aligned with SMCC's core values.

An International study shows that gamification and role-playing games (RPGs) are great learning aids for increasing engagement, motivation, and moral reasoning. Kim and Castelli [1] demonstrated that game-based learning enhances critical skills such as decision-making and teamwork. Oliveira et al. [2] also observed that students' experiences with gamified learning systems vary based on their behavioral and demographic profiles, suggesting the need for

adaptable and inclusive educational games. These studies provide compelling evidence that combining storytelling, moral decision-making, and game mechanics can create immersive learning experiences that foster character development.

In the local, Filipino-made tabletop RPGs highlighting national mythology and folklore have gained recognition as educational and cultural tools [3]. These games have demonstrated that storytelling paired with interactivity can increase Filipino kids' cultural knowledge and ethical consciousness. However, the majority of existing local educational games are concerned with artistic or linguistic content rather than the establishment of institutional values. This study presents Michaelinian Quest, a narrative-driven role-playing game (RPG) that encourages students to explore and apply fundamental Catholic and Michaelinian ideals. The game fosters moral decision-making rooted in students' real-life experiences.

This research introduces Michaelinian Quest, a role-playing game (RPG) that helps students understand and embody SMCC's core values. The game promotes Catholic and Michaelinian ideals through interactive storytelling and gaming. It empowers users to make moral decisions based on personal experiences. The approach tackles the problem of low student engagement with traditional methods of teaching values. It translates abstract moral concepts into engaging, decision-based scenarios. In the game, players take on the role of students navigating campus-inspired adventures, with their decisions and actions representing the following values: Socially Responsible, Missionary, Committed, and Competent. Through mission progression and real-time feedback, the game promotes moral reasoning, collaboration, and

leadership. The success of gamification is relative to the engagement of the learner/player, which game mechanics and aesthetics primarily enhance [4].

Michaelinian Quest has a digital rewards system pays players with certificates and badges for achieving value-based goals. The game's framework supports continuous reflection and application of ideas, connecting the classroom learning to engage in digital activities. Using Gamification concepts, the system creates a platform that mixes enjoyment and education, enabling Participants to experience moral growth through guided decision-making and missions that require collaboration.

This study presents an innovative teaching method that uses technology to promote value formation through engaging learning experiences. It benefits students by making moral training more engaging and relevant to their everyday lives, and it helps instructors and administrators by providing an additional platform for values integration. Furthermore, it aligns with SMCC's Core Values of developing socially responsible and competent individuals anchored in missionaries and Quality Education by encouraging inclusive, equitable, and meaningful learning opportunities through digital innovation.

1.2Literature Review

This chapter represents the related literature and study to expand on these findings by using RPG mechanics and narrative immersion to promote the Michaelinian identity and core values, which are socially responsible, missionary, competent, and committed in a more

engaging, technology-driven manner. The related literature comes from electronic materials, e-books, websites, scholarly articles, and journals.

Game-Based Learning and Educational Gamification

Gamification and Game-Based Learning (GBL) are innovative educational strategies incorporating play elements to enhance student engagement and learning outcomes. Game-based learning is considered one of the most innovative approaches to increase students' interest in education by playing games[5]. It converts traditional lessons into engaging experiences that foster participation, persistence, and enjoyment. In contrast, game-based learning employs actual digital or physical games specifically intended to meet educational goals. It refers to using games as an educational tool or strategy to facilitate learning and engagement[6].

Digital games significantly improve engagement, motivation, and challenge in educational environments. Motivation increases as students receive immediate feedback, achieve rewards, and experience a sense of progress and accomplishment, fulfilling their intrinsic need for competence and growth. Many researchers show that digital game-based learning is becoming a powerful tool in education, making learning more enjoyable, easier, and efficient[7].

Gamification and Value Formation

Gamification has emerged as a powerful pedagogical approach integrating game design elements such as points, badges, levels, and rewards into non-game learning environments to increase engagement and motivation. Recent studies highlight that their excitement and motivation were apparent. Using various tools and media, learning through play is child-centered

and fosters joy[8]. Through game-like challenges, learners experience moral and social scenarios reinforcing values such as honesty, teamwork, and perseverance. Students are now more active and motivated to use the different media teachers provide to create new projects. Research supports this, indicating that well-designed games create an enjoyable and stimulating learning environment that enhances children's intrinsic motivation to participate [9].

Digital Inclusiveness and Institutional Support

Digital inclusion ensures all individuals can access technology regardless of socioeconomic background, geographic location, or technical aptitude. Everyone has equal access to digital technology and educational possibilities. The prior research supports the idea that play-based activities enhance students' participation and engagement when complemented by diverse media and sufficient teacher support [10]. In education, the emphasis is on bridging the digital divide so that every learner can participate fully in digital learning environments. According to Aljarallah and Lock [11], digital inclusion involves not only access to technology but also the development of digital literacy, affordability, and accessibility of online resources. This inclusiveness enhances participation, engagement, and equity, ensuring that every student actively participates in the transition to technology-driven learning and is not disadvantaged. Studies also suggest that corporate social responsibility (CSR) initiatives can help address funding shortages by providing subsidized educational games[12]. Promoting digital inclusiveness in education requires collective efforts from institutions, policymakers, and the private sector to

create accessible, affordable, and literacy-driven digital ecosystems that empower all learners to thrive in the modern educational environment.

Adoption and Challenges in Game-Based Learning

Game-Based Learning (GBL) in education has grown significantly in recent years due to its potential to enhance motivation, engagement, and learning outcomes. This method relies on the premise that games have an intrinsic appeal, making learning more enjoyable and immersive[13]. However, its implementation continues to face several challenges. Studies reveal that the most common barriers include inadequate infrastructure, insufficient teacher training, and a lack of institutional support, which hinder educators from effectively integrating GBL into their curricula [14]. Teachers often find it challenging to align games with learning outcomes, manage classroom dynamics, and assess student performance in game environments. Games fulfil three basic psychological need: autonomy, competence, and relatedness,game structure should be implemented properly to cover all three[15]. While GBL presents innovative opportunities for education, its success depends on addressing these barriers through institutional investment, teacher training, and inclusive digital infrastructure that supports educators and learners. Holistic and integrated approaches such as game-based curricula and the Reggio Emilia approach have proven effective in promoting active learning and engagement[16].

1.3 Project Objectives

The main goal of this study is to create a story-driven role-playing game called Michaelinian Quest. This game will promote the key values of Saint Michael College of Caraga through interactive storytelling, fun gameplay, and a digital reward system that includes certificates and badges.

Specifically, it aims to:

1. To develop an interactive role-playing game that integrates narrative-driven quests aligned with the Michaelinian core values.

2. Design a functional and interactive role-playing game that fosters Socially Responsible, Missionary, Committed, and Competent among students.

3. Implement a digital rewards system with certificates and badges to enhance motivation and participation.

1.4 Scope and Limitation

The Project Michaelinian Quest is a browser-based educational role-playing game designed to teach Saint Michael College of Caraga's fundamental ideals of Socially Responsible, Missionary, Commitment, and Competence. Set in the local environment of Nasipit, Agusan del Norte, the initiative reflects the ideals and educational mission of Saint Michael College of Caraga. Through narrative-driven missions and decision-based gameplay, the game promotes moral reasoning and value-based decisions that align with these institutional goals. It was created with RPG Maker

MV and deployed through Netlify. It is accessible via web browsers, making it convenient for students who use ordinary desktop or laptop devices.

While the game focuses on educational impact over visual fidelity, it employs simple 2D visuals appropriate to its aim and the school's available resources. Basic visuals and advanced gameplay features, such as user accounts and progress tracking are limited. The initial development and testing will occur in elementary, junior high, senior high, and college environments. The Review will focus on user experience and how well the game develops an understanding and application of the college's fundamental values, laying the groundwork for future enhancements. However, the game is only available on a PC, not mobile, iOS, or Apple devices.

1.5 Definition of Terms

RPG Maker MV – A 2D game development engine that creates role-playing games. It uses JavaScript as its scripting language and features a map editor, eventing system, and plugin support. RPG Maker MV supports desktop and web-based games (HTML5), unlike older versions.

JavaScript – A programming language used in RPG Maker MV to create plugins and customize the game beyond the default settings. Developers can write JavaScript code to modify combat, menus, systems, etc.

Plugin – A file written in JavaScript that extends or changes RPG Maker MV's core functionality allows users to place plugins in the game's js/plugins folder. At the same time, the Plugin Manager helps manage them. These plugins enable features like quest systems, crafting, or custom HUDs.

Event – A core feature of RPG Maker MV handles in-game interactions like dialogues, cutscenes, teleportation, and item collection. Developers place events on maps and control them through event commands and conditions.

NPC (Non-Playable Character) – Characters that populate the game world but players do not control often serve to provide dialogue, quests, or environmental storytelling.

Map – An area in the game world designed using tiles. Maps are built in the editor and can include events, characters, and environmental features.

Tileset – A collection of tiles (graphics) used to build maps in RPG Maker MV. Each map is assigned a tileset, which includes terrain, walls, decorations, etc.

Switches and Variables – In RPG Maker MV, logic tools can control events using on/off (true/false) values. For instance, players can track if they have defeated a boss. Additionally, they can use variables to store numbers or text, which helps monitor player stats, decisions, or progress, such as how much gold they have collected or how many enemies they have defeated.

Branching Dialogue – A system that allows conversations to change based on player choices. Adding interactivity can influence story outcomes or character relationships.

Plugin Manager – The tool inside RPG Maker MV, used to manage which JavaScript plugins are active in a game, allows for enabling, turning on, or turning off plugins.

UI (User Interface) – The on-screen menus and information the player interacts with, such as health bars, inventory screens, or dialogue boxes. RPG Maker MV allows developers to customize UI using JavaScript or plugins.

CHAPTER II
METHODOLOGY

2.1 Research Design

This project will implement a Design and Development Research approach to create Michaelinian Quest. This story-driven role-playing game promotes core school values such as being Socially Responsible, Committed, and Competent. The structure ensures the game is pedagogically sound and engaging through iterative design, development, and evaluation. These cycles enable the game to evolve based on feedback from students and educators, ensuring its relevance and effectiveness in shaping values.

2.2 Software Development Life Cycle

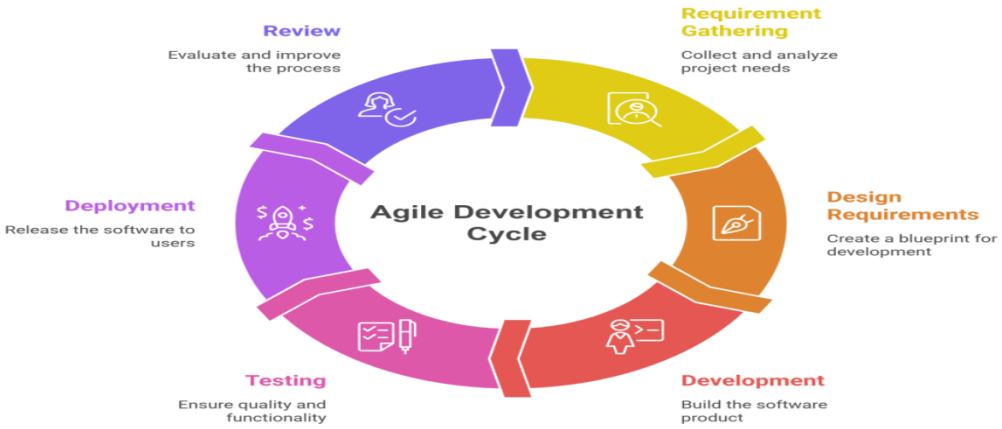


Figure 1. Agile Methodology

Figure 1: Agile Methodology will be used to develop Michaelinian Quest to ensure it is learner-centered, adaptable, and responsive to the needs of both students and teachers. Agile

development breaks the game into small, playable segments called sprints—such as character creation, quest design, and value-based decision-making scenarios. Each sprint produces a functional part of the game that can be tested and refined before proceeding. This strategy allows for ongoing improvement based on feedback from the classroom. It tackles issues like digital literacy, engagement, and narrative clarity. Michaelian Quest can turn into a useful and active tool for values education with short, focused development cycles.

2.3 System Architecture

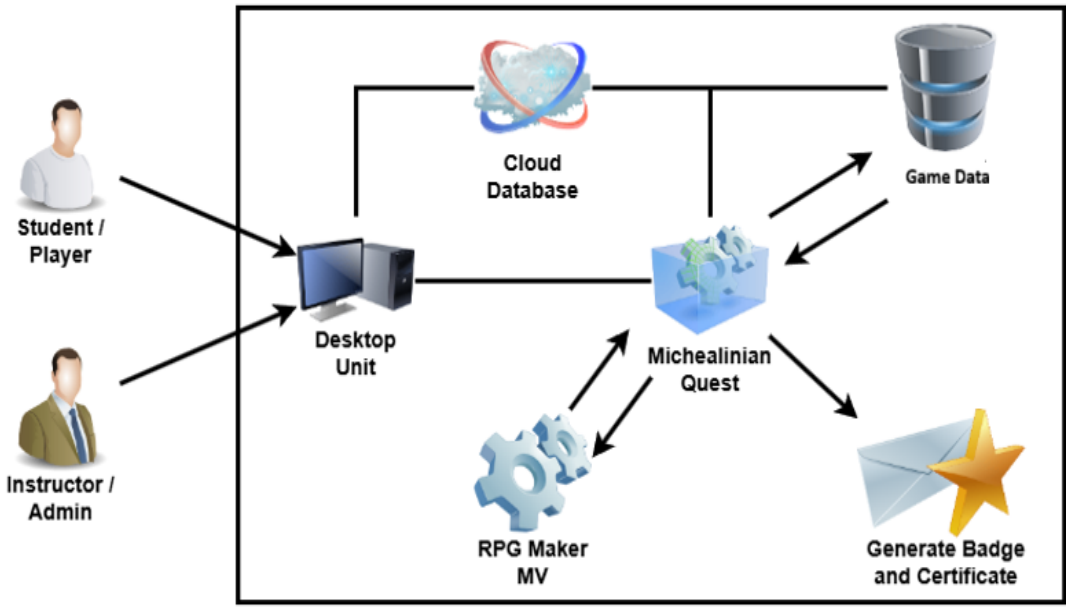


Figure 2. System Architecture

Figure 2 shows the system architecture for Michaelinian Quest, a browser-based educational role-playing game. Students or gamers access the game through a web browser using services like Netlify on desktop or laptop computers. The instructor or Administrator uses RPG Maker MV to design and update the game material offline, then redeploys the modified game to the hosting platform. The system focuses on static deployment. Badges and certificates are displayed based on player performance and in-game decisions, reinforcing the essential ideals of Saint Michael College of Caraga.

2.4 Conceptual Framework

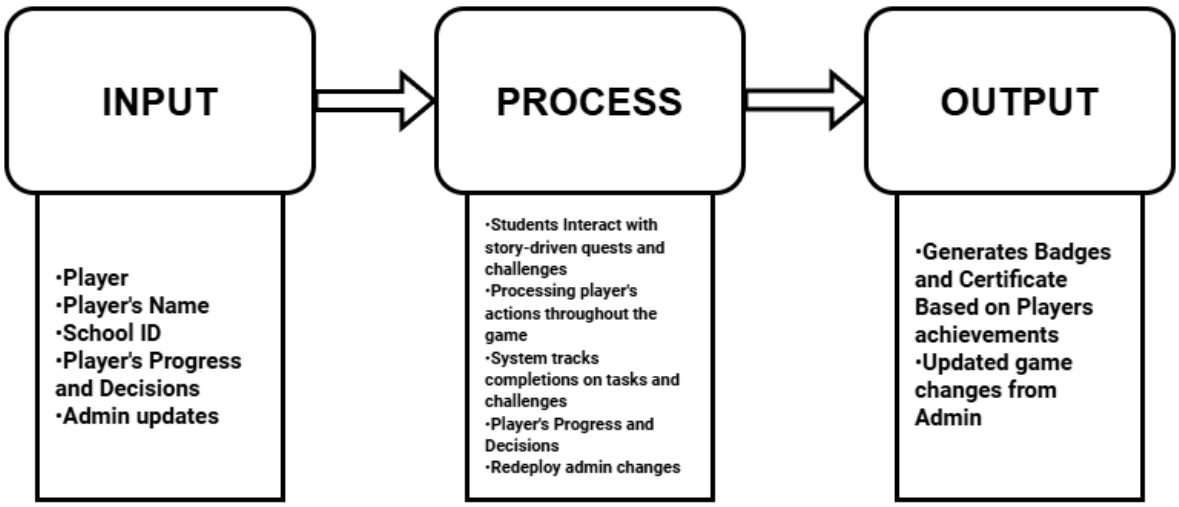


Figure 3. Input-Process-Output Diagram of Michaelinian Quest

The Diagram shows Michaelinian Quest's Input-Process-Output (IPO) paradigm, which describes the flow of information and interaction during gameplay. In the Input stage, players

enter basic information such as their name, which is then utilized in-game for personalization. Using built-in save facilities, the game may also save the player's choices and progress locally. The Process stage provides a significant gaming experience. In this stage, players take part in story-driven quests, make value-based choices, and complete tasks that reflect the essential ideals of Saint Michael College of Caraga. The system records the player's in-game actions and tracks task completions within the session or save file. In the Output stage, players receive recognition for their achievements through in-game badges and certificates. This acts as a reward for completing tasks and embodying the promoted values. The IPO diagram shows how player input creates meaningful in-game outcomes through interactive storytelling and moral decision-making.

2.5 Use Case Diagram



Figure 4. Michaelinian Quest Use Case

Figure 4 illustrates the interaction between two key users—Player and Admin—and the system functions. After completing the game, players will receive a certificate. The Administrator manages the game's data and makes adjustments to keep it running well and to improve its performance. This use case shows how each user's role and level of access help create a smooth gaming experience and effective game management.

2.6 Activity Diagram

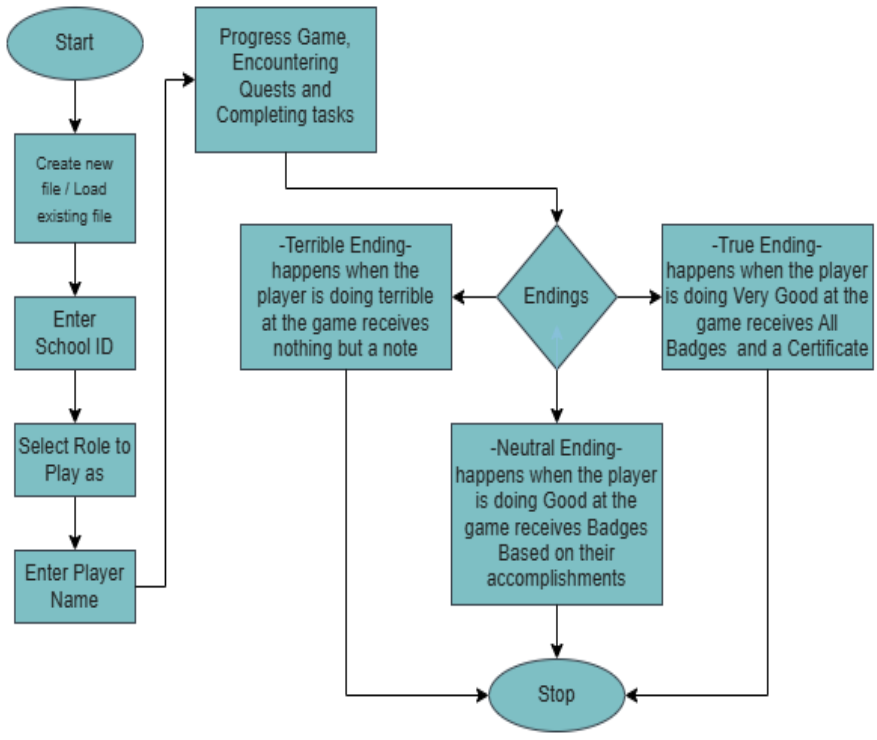


Figure 5 Michaelinian Quest Activity Diagram (Student/Player)

Figure 5 illustrates the workflow of Michaelinian Quest, starting with the player launching the game and creating or loading a Save file. The player then enters their school ID, selects their role, and chooses their player name. Once the player finalizes these settings, they can progress through the game, completing tasks and encountering quests.. At the end of the game, the player's performance will determine which of the three possible endings they will receive: the "Terrible Ending," "Neutral Ending," or "True Ending".



Figure 6. Michaelinian Quest Activity Diagram (Instructor / Admin)

Figure 6 illustrates the development pipeline for the Michaelinian Quest game. The technique begins with acquiring game data, including objects, scripts, and player progress. Next, the designers use RPG Maker MV, a game development tool, to make the necessary tweaks and improvements. After making the updates, we redeploy the game using GitHub, ensuring users can access the most recent version. This cycle continues as we implement additional adjustments and revisions to enhance playability and performance.

2.7 Sequence Diagram

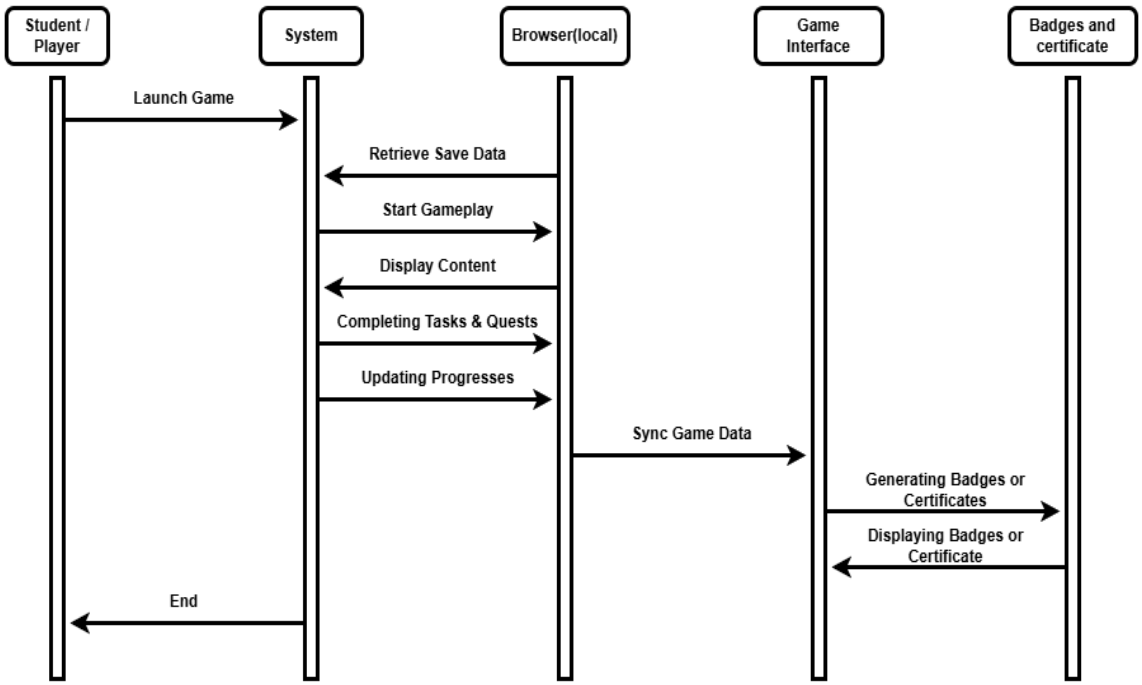


Figure 7. Sequence Diagram (Student/Player)

Figure 7 illustrates a series of interactions between the Student and Player with the game system once they launch it in the browser. The procedure begins. When a student or participant starts the game in their browser. The system retrieves saved data from local storage or starts a new session. The player interacts with the game interface by completing tasks and making decisions during the game. As the player moves forward, their game data updates and syncs with the local storage. When the player reaches specific milestones, the system generates a badge and certificate, which are displayed to the player to recognize their achievements. Once the game ends, the player can view their rewards and performance results.

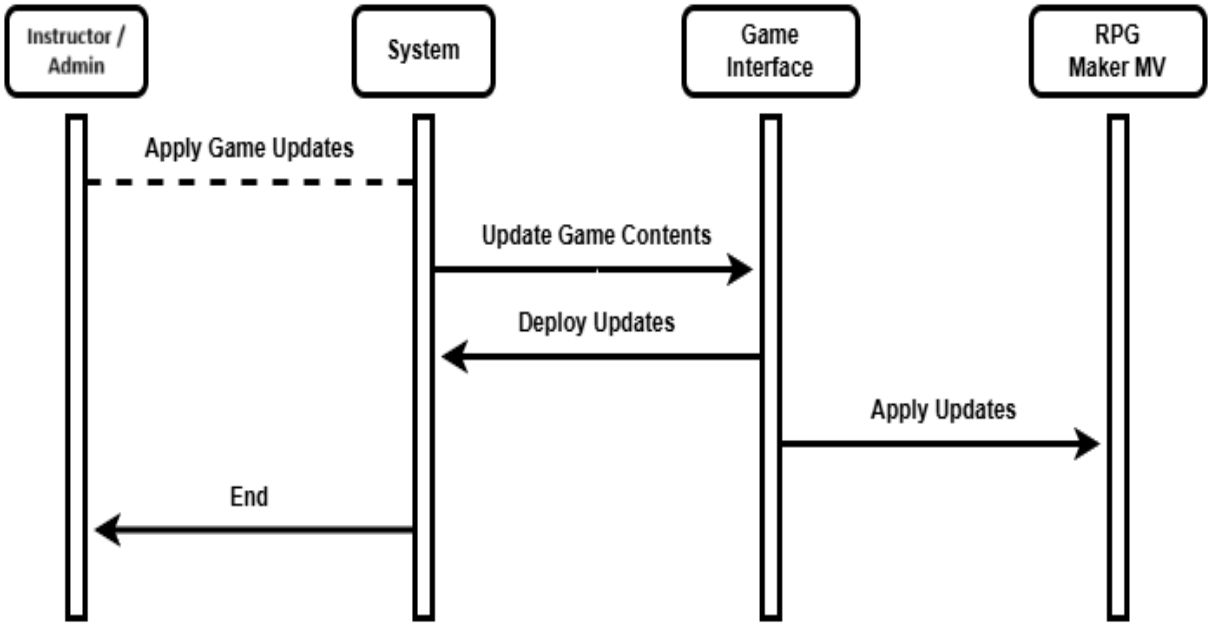


Figure 8. Sequence Diagram (Instructor/Admin)

Figure 8 illustrates the Instructor/Admin's role in managing the game, where their main task is to update the game content using RPG Maker MV. After applying changes, the admin redeploys these updates to the live game.

2.8 ERD Diagram

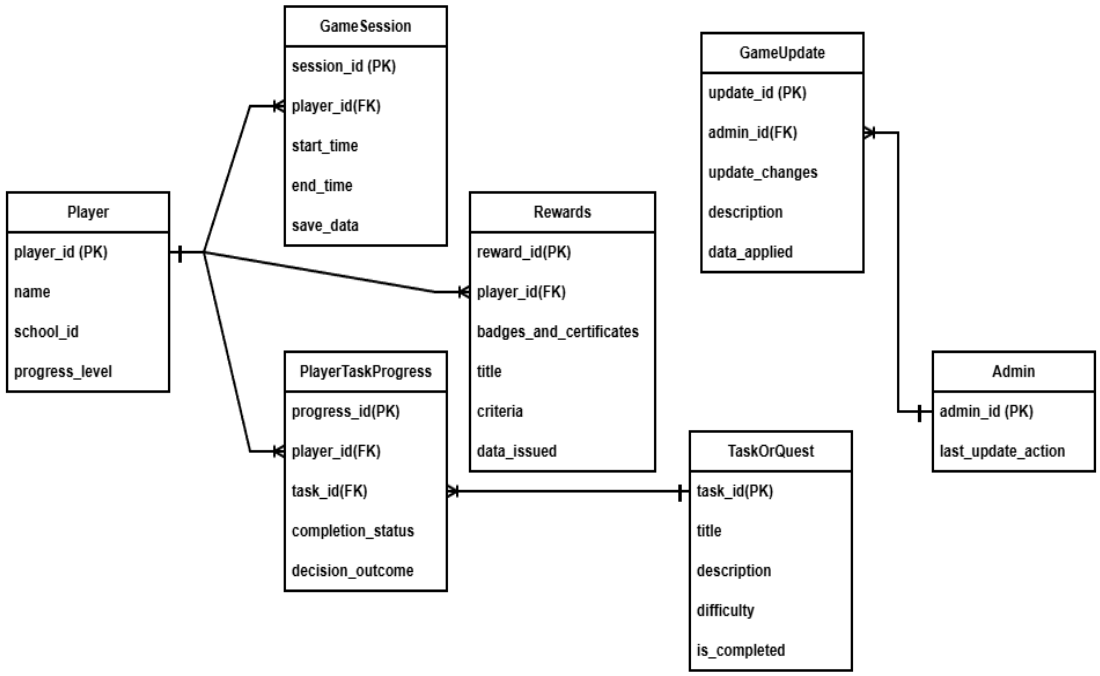


Figure 9. Entity Relationship Diagram

Figure 9 shows the Entity Relationship Diagram (ERD) of the Michaelinian Quest platform. It illustrates the structure of its main entities and their connections. At the center of the system is the Player entity, which has player_id as the primary key and a name attribute. Each player can take part in multiple quests. This creates one-to-many relationships between Player and Quest. The diagram explains how quests track activities and decisions. It rewards players and forms the base of Michaelinian Quest's gameplay structure.

2.9 User Interface Design

Figure 10. User Interface Design



Figure 10 illustrates Michaelinian Quest's user interface, namely the title screen arrangement. The UI is basic and user-friendly, with clickable buttons that let the player start a new game, continue from a previous save file, or exit the application. The title screen is the player's primary access point to the game, ensuring easy navigation and a smooth transition into gameplay. Its minimalist design emphasizes functionality and clarity, allowing players to get right into the game.

Figure 11. User Interface Design



Figure 11 illustrates the user interface, which requires players to provide their school ID and player name before beginning the game. This method is part of the initial setup and allows the system to recognize and personalize the player's progress. The straightforward and structured interface makes it easy to use, taking players through the registration process quickly before starting the primary gameplay.

Figure 12. User Interface Design



Figure 12 shows the role selection screen, which allows participants to choose their category by moving left or right. The available roles correspond to their academic level: elementary, high school, and college. This function guarantees that each player is assigned the right game path based on their selected role, tailoring the gameplay experience to their educational level.

Figure 13. User Interface Design



Figure 13 illustrates the user interface, which requires players to enter their names before starting the game. Like the school ID input screen, this step personalizes the gameplay experience and allows the system to link progress and achievements to the player's profile. The uncomplicated layout ensures a rapid, clear, and straightforward entry process.

Figure 14. User Interface Design



Figure 14 shows the point at which actual gameplay begins, depending on the role selected by the player. Following the setup phase, players take assigned positions in their respective games. This feature provides a personalized gameplay experience, ensuring that each player chooses the path best suited to their position and goals in the game.

Figure 15. User Interface Design

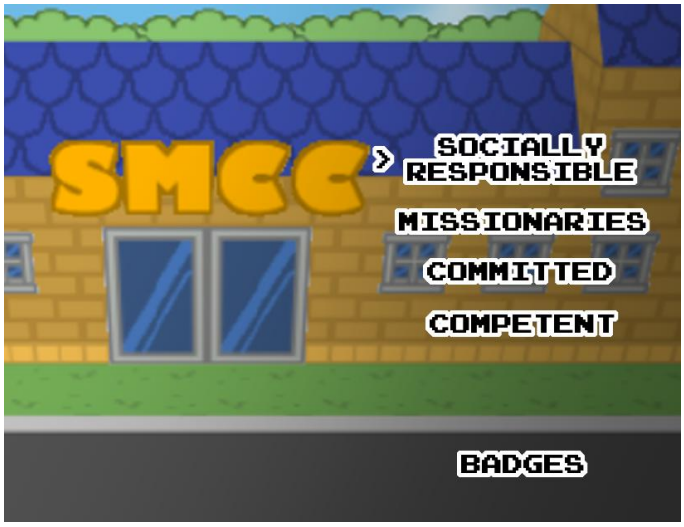


Figure 15 shows the UI of the game after the prologue ends, it shows 5 choices in which the students play based on what core it is. Upon pressing a choice button it rolls up to 5 to pick a scene to play, this is for the players to not play the same scene over and over again

Michaelinian Quest

Prologue / beginning of the game
Player's name default: Miguel/malecharacter, Mikha/femalecharacter (can be customized)
A new student at smcc

(the story can be changed depending on what role the student is playing)



at school Fr. Dominic welcomes the new students in the school

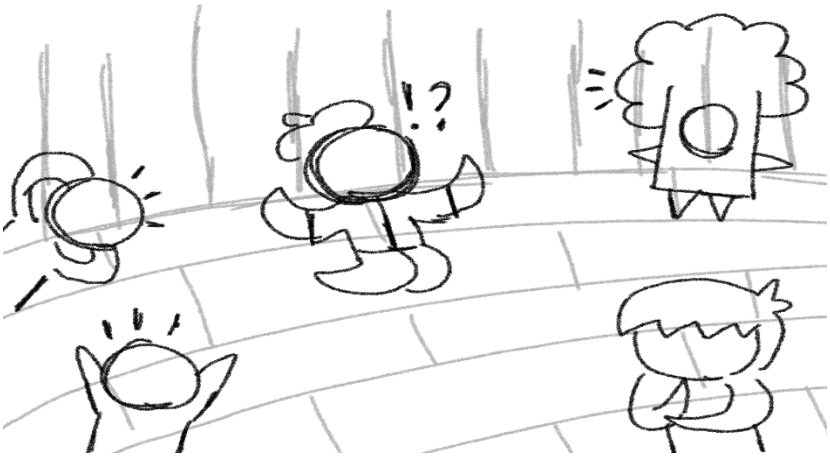
Scene:
mother says goodbye to the protagonist
"Study well, my child. Be kind, and be helpful to others in need"
at school Fr. Dominic welcomes the new students in the school
Example dialogues: "To be a Michaelinian is to be the light for others"
more dialogues and scenes for the day



Chapter1 = Socially Responsible
Scene:
An activity clean-up Drive at nasipit boulevard, The class joins a coastal clean-up(some student don't want to help)
Choices:
a. help and encourage others to help also (+2 points)
b. help quietly don't make contact from other schoolmates (+1 points)
c. skip cleaning and don't help at all (-2 points)
Result:
>Teacher will congratulate you for being helpful
>Teacher will say "Good job! You care for others"
>You learn that small actions help the whole community

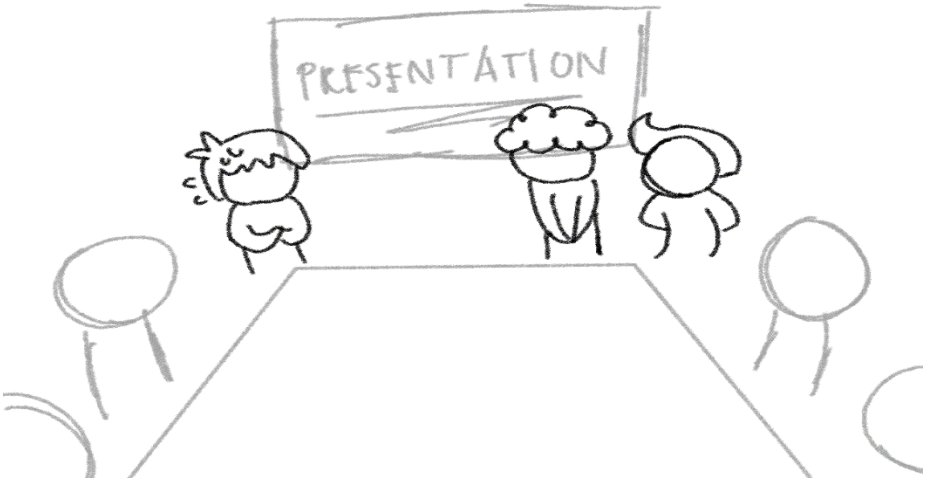


Chapter2 = Missionary
Scene:
An outreach in barangay 3 Flood victims need of help. Fr Dominic invites volunteers
Choices:
a. Comfort and help the people in need (+3 points)
b. giving goods to the people and avoid interactions (+1 points)
c. refuse to join (-3 points)
Result:
>You learn that sharing hope is part of your faith



Chapter3 = Committed
Scene:
Lost Script before the Play, Your Group loses the script for a school play but you have a copy at home
Choices:
a. go home and get the copy for the play (+3 points)

b. ignore it and let the group find and struggle a way (+1 points)
c. skip practice at all (-2 points)
Result:
>The play succeeds. You learn that commitment means doing your part.



Chapter4 = Competent
Scene:
Presentation Day Problem, the laptop crashes before your presentation
Choices:
a. gather up your courage and present it on memory (+3 points)
b. cowered out and blame your groupmates (-1 points)
c. no choice but to give up (-2 points)
Result:
>Teacher praises your calm and skills.
>You learn that competence means doing your best, even in difficult situations



Final Chapter = The boulevard
Scene:
You've reach the ending and walk by the sea at the sunset and Fr. Dominic meets you

Ending 1 if high points:
"You lived as a True Michaelinian – faithful and kind"

Ending 2 if medium points:
"You're still learning, but your heart grows brighter"

Ending 3 if low points:
"Even if we fail, God's light waits for us to return"

you smile and watch the waves

processes the players actions throughout the game
prints certificate and badges depending on the players performance

Ending credits

2.10 Software Platforms, Development Environments and Tools

Component	Specification	Usage

RPG Maker (Software)	RPG Maker MV	Core game development platform. Used for creating game mechanics, mapping, scripting events, and creating game content.
Graphics Software	Adobe Photoshop, Clip Studio Paint, Roblox Studio, Blender	Used for creating and editing sprites, tilesets, backgrounds, and other visual assets.
Programming Languages	JavaScript (for RPG Maker MV)	Used for customizing gameplay features, adding complex scripts, or enhancing game functionality beyond the default RPG Maker capabilities.
Database Management	RPG Maker built-in database (also can be supplemented by external databases or JSON files)	For managing the game's items, characters, enemies, skills, and more. The database allows easy access to game content and quick changes.

Table 2
Hardware Requirements

COMPONENTS	SPECIFICATION	USAGE
Processor	Intel Core i5 or equivalent (or AMD Ryzen 5)	Ideal for smoother development, testing larger games, and running complex scripts and events.

RAM	8 GB RAM or higher	Ideal for more demanding projects with large assets or more complex scripts.
Graphics Card	NVIDIA GeForce GTX 1050 or equivalent, or AMD Radeon RX 560	Best for high-quality visual assets and ensuring that the game's graphical elements are displayed correctly and smoothly.
Storagess	1 GB free hard drive space minimum	Required to install RPG Maker and run the game. A larger project will need more space for assets and game builds.

System Evaluation

This section presents the evaluation procedures conducted to assess the Michaelinian Quest system developed using RPG Maker. The evaluation focuses on measuring the system's functional suitability, performance efficiency, and usability. Its purpose is to determine the game's impact on players' values, appreciation, engagement, and learning outcomes. A four-point rating scale was used, with 4 as the highest and 1 as the lowest.

Data were analyzed and interpreted based on the following parameters:

Table 3

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional

3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate a system's functional suitability according to the ISO survey.

Table 4
Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to the usability of the system.

Table 6

Respondent's Distribution

Position	**N(Population)**	**Percentage**
Player(Student)	30	55.56%
Admin(Instructor)	1	11.11%
Total:	6	

Their assessments give useful information regarding the usability, effectiveness, and overall efficiency of the system. On another hand, students' opinions bring to light how the platform connects with their learning needs, engagement levels, and overall user experience.

2.13 Ethical Standards.

Michaelinian Quest: A Story-Driven Role-Playing Game Exploring Saint Michael College of Caraga's Core Values was created and implemented strictly with recognized ethical standards for research, game production, and participant participation. Throughout the design, testing, and evaluation process, we uphold honesty, integrity, and regard for human dignity.

A.Intellectual Property Rights (IPRs)

All assets, including images, music, voice, and programming scripts, are originals, licensed, or used with proper permission and attribution. The researchers ensure strict respect for copyright laws and institutional intellectual property policies.

B. Informed Consent

Before participating in the game testing and evaluation, we fully inform participants of the study's aims, processes, and their rights. Written or digital consent is given voluntarily, with the understanding that individuals can withdraw at any moment without penalty.

C. Data Privacy and Confidentiality

All personal information provided by participants is kept strictly confidential. Data is anonymize and securely stored to prevent unwanted access, per the Data Privacy Act of 2012 (Republic Act No. 10173).

D. Voluntary Participation

Participation in all phases of the study is entirely voluntary. Researchers do not use compulsion, manipulation, or undue influence to encourage participation, and individuals can drop out at any time.

E. Minimization of Harm

The researchers take all essential procedures to guarantee that participants are not physically, emotionally, or psychologically harmed during gameplay or data collection. The game's content encourages moral thinking without causing distress or discomfort.

F. Beneficence

The project aims to positively influence the academic community by encouraging values education through interactive digital learning. Participants enjoy engaging gameplay experiences that foster ethical and moral decision-making.

G. Just and Fair Participant Selection

Participant selection is fair and free from discrimination. Everyone has an equal chance to take part, no matter their gender, financial situation, or academic performance.

H. Data Integrity and Accuracy

All acquired data is accurately documented, reviewed, and reported. The researchers make sure that their findings are not faked, manipulated, or misrepresented in any manner.

I. Transparency in Reporting

The study presents its findings, methods, and limitations honestly and clearly. The researchers fully disclose any potential conflicts of interest and outside influences.

J. Use of Patented or Copyrighted Materials

All materials used in the game, including software tools, design assets, and academic references, are properly credited and used in compliance with applicable copyright and licensing agreements.

K. Ethical Considerations for Human Participation

All researchers treat participants with dignity and fairness during the study. The researchers adhere to the ethical research protocols set by Saint Michael College of Caraga and relevant research ethics standards.

L. Responsible Use of AI and Related Technologies

When developers use artificial intelligence or automated technologies in game design or data analysis, they must manage them responsibly to ensure fairness, openness, and the absence of bias and unethical data manipulation.

M. Ethical Approval

The ethical approval application form for Michaelinian Quest seeks authorization to conduct the study in compliance with Saint Michael College of Caraga's ethical standards, ensuring informed consent, confidentiality, and the responsible treatment of all participants.

2.14 Data Gathering Procedure

Michaelinian Quest: A story-driven role-playing game that explores the core values of Saint Michael College of Caraga, carefully collected data to ensure accuracy and ethical compliance. The researchers got approval and asked for permission to study selected students and faculty members at SMCC. They used different research tools to evaluate the game's usability, engagement, and instructional value. Researchers collected and analyzed the data using descriptive statistics to determine the game's effectiveness in promoting SMCC's core values: socially responsible, missionary, committed, and competent. They upheld ethical standards such as informed consent, confidentiality, and voluntary participation throughout the process.

REFERENCE

[1] J. Kim and D. M. Castelli, “Effects of gamification on behavioral change in education: A meta-analysis,” Int. J. Environ. Res. Public Health, vol. 18, no. 7, p. 3550, 2021. [Online]. Available: https://doi.org/10.3390/ijerph18073550

[2] W. Oliveira, A. Toda, P. Toledo, L. Shi, J. Vassileva, I. I. Bittencourt, and S. Isotani, “Does tailoring gamified educational systems matter? The impact on students’ flow experience,” in Proc.

53rd Hawaii Int. Conf. System Sciences, 2020, pp. 1226–1235. [Online]. Available: https://hdl.handle.net/10125/63891

[3] N. E. Correa, A. Garcia, and M. G. Patino, "The use of narrative role-playing games in communication skills development among Filipino youth in Metro Manila," Unpublished Manuscript, 2021. [Online]. Available: https://scholar.google.com/

[4] M.Kalogiannakis, S.Kalogiannakis, A.-I. Zourmpakis, Gamification in Science Education. A Systematic Review of the Literature. Educ. Sci. 2021, [Online] Available:https://doi.org/10.3390/educsci11010022

[5] M. Priyaadharshini, R. Dakshina, and S. Sandhya, "Learning analytics: Game-based learning for programming courses in higher education," Procedia Comput. Sci., vol. 172, pp. 468–472, 2020. [Online]. Available: https://doi.org/10.1016/j.procs.2020.05.143

[6] F. Y. Li, G. J. Hwang, P. Y. Chen, and Y. J. Lin, "Effects of a concept mapping-based two-tier test strategy on students' digital game-based learning performances and behavioral patterns," Comput. Educ., vol. 173, p. 104293, 2021. [Online]. Available: https://doi.org/10.1016/j.compedu.2021.104293

[7] M. Hafeez, "Effects of game-based learning in comparison to traditional learning to provide an effective learning environment: A comparative review," Contemp. Educ. Res. J., vol. 12, no. 2, pp. 89–105, 2022. [Online]. Available: https://doi.org/10.18844/cerj.v12i2.6374

[8] T. E. Coleman and A. G. Money, “Student-centred digital game–based learning: A conceptual framework and survey of the state of art,” Higher Educ., vol. 79, no. 3, pp. 415–457, Jul. 2020. [Online]. Available: https://doi.org/10.1007/s10734-019-00417-0

[9] M. Arztmann, L. Hornstra, J. Jeuring, and L. Kester, “Effects of games in STEM education: A meta-analysis on the moderating role of student background characteristics,” Stud. Sci. Educ., vol. 59, no. 1, pp. 109–145, 2023. [Online]. Available: https://doi.org/10.1080/03057267.2022.2057732

[10] H. J. Bang, L. Li, and K. E. Flynn, “Efficacy of an adaptive game-based math learning app to support personalized learning and improve early elementary school students’ learning,” Early Child. Educ. J., vol. 51, no. 4, pp. 717–732, Apr. 2023. [Online]. Available: https://doi.org/10.1007/s10643-022-01332-3

[11] S. Aljarallah and R. Lock, “An investigation into sustainable e-government in Saudi Arabia,” Electron. J. e-Government, vol. 18, no. 1, Apr. 2020. [Online]. Available: https://doi.org/10.34190/EJEG.18.1.001

[12] J. Vlassis, A. Baye, A. Auquière, A. F. de Chambrier, C. Dierendonck, N. Giauque, S. Kerger, C. Luxembourger, D. Poncelet, M. Tinnes-Vigne, Y. Tazouti, and A. Fagnant, “Developing arithmetic skills in kindergarten through a game-based approach: A major issue for learners and a challenge for teachers,” Int. J. Early Years Educ., vol. 31, no. 2, pp. 419–434, 2023. [Online]. Available: https://doi.org/10.1080/09669760.2022.2138740

[13] H. Yaman, C. Sousa, P. P. Neves, and F. Luz, “Implementation of game-based learning in educational contexts: Challenges and intervention strategies,” Electron. J. e-Learning, vol. 22, no. 1, p. 15, 2024. [Online]. Available: https://doi.org/10.34190/ejel.22.10.340

[14] P. Saleme, T. Dietrich, B. Pang, and J. Parkinson, “Design of a digital game intervention to promote socio-emotional skills and prosocial behavior in children,” Multimodal Technol. Interact., vol. 5, no. 10, p. 58, 2021. [Online]. Available: https://doi.org/10.3390/mti5100058

[15] A.Manzano-León, P. Guerrero, M.A.; Guerrero-Puerta, L.; Aguilar-Parra, J.M.; Trigueros, R.; Alias, A. Between Level Up and Game Over: A Systematic Literature Review of Gamification in Education. Sustainability 2021, 13, 2247. [Online] Available:https://doi.org/10.3390/su13042247

[16] J.Krath, L.Schürmann, & H. F. O.von Korflesch, Revealing the theoretical basis Hibana, et. al. (2024) Exploring the role of game-based learning in early childhood of gamification: A systematic review and analysis of theory in research on gamification, serious games and game-based learning. Computers in Human Behavior, 125(July), 106963. 2021, [Online] Available: https://doi.org/10.1016/j.chb.2021.106963

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ORGANIZATION	POSITIONS HELO
N/A	N/A

